

ROSHNI

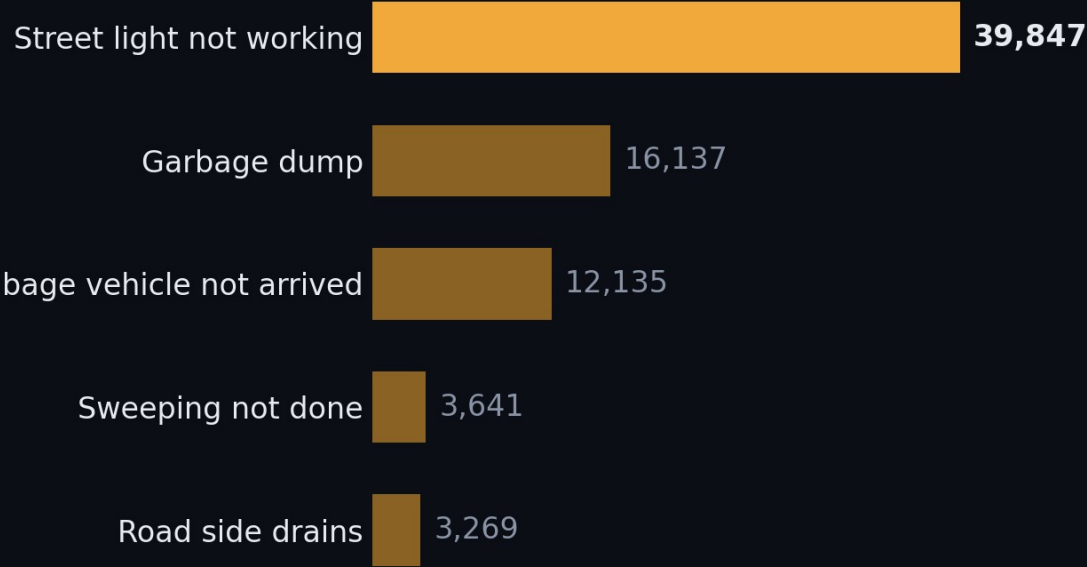
The streetlight repair queue, reordered by who is actually walking.

BENGALURU'S MOST-FILED COMPLAINT

31.4%

39,847 of the 126,974 civic complaints Bengaluru filed between 1 January and 19 June 2025 were “Street Light Not Working” — about 236 a day.

The largest complaint category in the city. Garbage is second, at 16,137.



Not failing. Handed the wrong order.

96.4% of electrical complaints closed



40,632 of 42,138. Road maintenance closes 58%. They are not slow and not under-resourced.

The queue is sorted by who complained first.

And complaining is not the same as being exposed



Newer, peripheral, more car-owning wards file most. Older, denser, more-walked wards file fewer — not because their lights work.

Same register. Left: Grievance Status field. Right: Sub Category = "Street Light Not Working", grouped by Ward Name, 198 wards.

Three surfaces, one atom

Queue municipal **THE HERO**

A ranked worklist under a repair budget.

Route citizen

Safer and shortest walking route, side by side, always both.

Report citizen

One tap: “this stretch is dark.”

The atom is the road segment, not the lamp. A pedestrian at 23:00 does not care whether a street has eleven lamps or fourteen — they care whether the next 200 metres is lit.

Darkness per segment is a quantity you can estimate. Lamp inventory is an enumeration you cannot obtain. BBMP does not have one either — their complaint records carry no coordinates and no pole IDs.

How it works

INPUTS

OSM walk graph

95,557 nodes · 98.4% connected

BMTG GTFS

6,187 night departures

Night-active POIs

3,417 of 9,199

Ward lamp density

421,113 lamps · 198 wards

OSM working = yes / no

1,128 labelled lamps

Citizen reports

GPS + timestamp

IUDX telemetry

gated — see slide 13

TWO MODELS

EXPOSURE

$E(\text{segment}) = \text{activity} + \text{structural}$

DARKNESS

$\text{Beta}(\alpha, \beta)$ per segment

mean = $P(\text{dark})$ · var = confidence

Estimated separately, from disjoint inputs.

That is what makes convergent validation possible.

FOUR OUTPUTS

Municipal queue

knapsack under budget + counterfactual

Safer route

weighted walk graph · both routes always

Confidence band

router refuses the least-certain fifth

Report targeting

upper confidence bound

Footfall enters the queue and the router with opposite signs — next slide.

Every element here exists in data/pipeline.py. No ML model, no cloud layer, no auth service.

We didn't pick these numbers. We measured them.

A citizen audit coded 469 Vasanthanagar lamps on two dates, 15 days apart, in 2019. That gives a measured repair rate and a measured failure rate — so the prior is calibrated, not chosen.

	2 July		
17 June	DEAD	OK	
DEAD	33	28	→ 45.9% of dead lamps repaired
OK	15	360	→ 4.0% of working lamps died

Equilibrium dead share = $0.040 / (0.040 + 0.459) = 8.0\%$

The prior it produces

```
P(lit) = coverage × 0.920
coverage = ward lamps/km ÷ 33.3
33.3 = 30 m standard pole spacing
0.920 = measured working rate
```

```
Beta( $\alpha_0$ ,  $\beta_0$ ) = (K·P(lit), K·(1-P(lit)))
K = 2, deliberately weak
```

Mean P(dark) across all 11,029 segments: **27.6%**

Two evidence weights worth naming: an OSM working=no tag moves β by 1.0; IUDX zero current after dusk would move it by 2.0. Everything decays on a 30-day half-life, so a repair clears a segment.

One variable, opposite signs

```
repair_priority = E · P(dark) · confidence      E multiplies  
route_penalty   = L · (1 + λd·P(dark) + λe/E + λu·var)  E divides
```

In the repair queue, footfall is **EXPOSURE**

More people at risk on a dark street, so fix it first.

In the router, footfall is **PROTECTION**

Eyes on the street, so prefer it. A bright empty road at 01:00 is not safe.

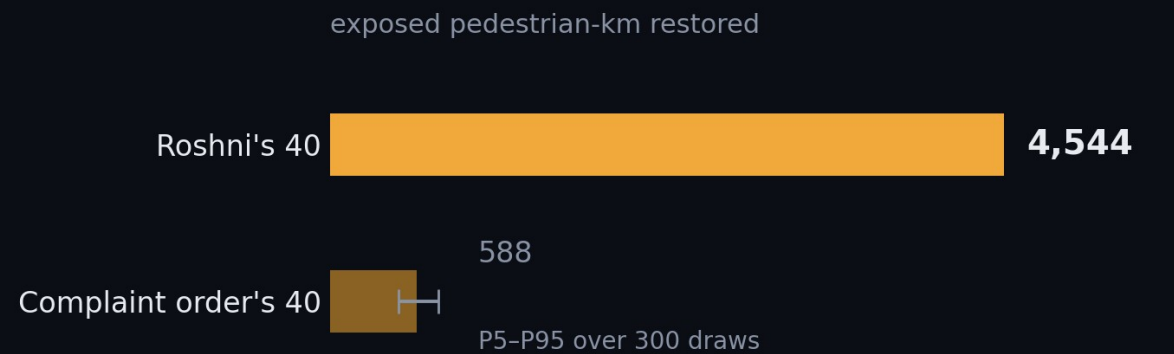
Same variable, opposite sign, depending on which surface consumes it. Use one sign in both and your router confidently sends a woman down an empty, brightly lit industrial road at one in the morning.

SAME CREW. SAME BUDGET. DIFFERENT ORDER.

7.7×

A crew can complete 40 repairs this week. Roshni's 40 restore 7.7× the exposed pedestrian-kilometres that the complaint queue's first 40 restore.

The gain is not from being cleverer inside a ward.
It is from visiting the right wards.



Stress-tested four ways, volunteered

vs the luckiest of 300 complaint-order draws	4.9×
vs a steelman BBMP picking the worst segment in every ward it visits	1.9×
restricted to only the wards where we hold lamp labels	6.4×
every lamp label deleted — ward prior × exposure alone	5.4×

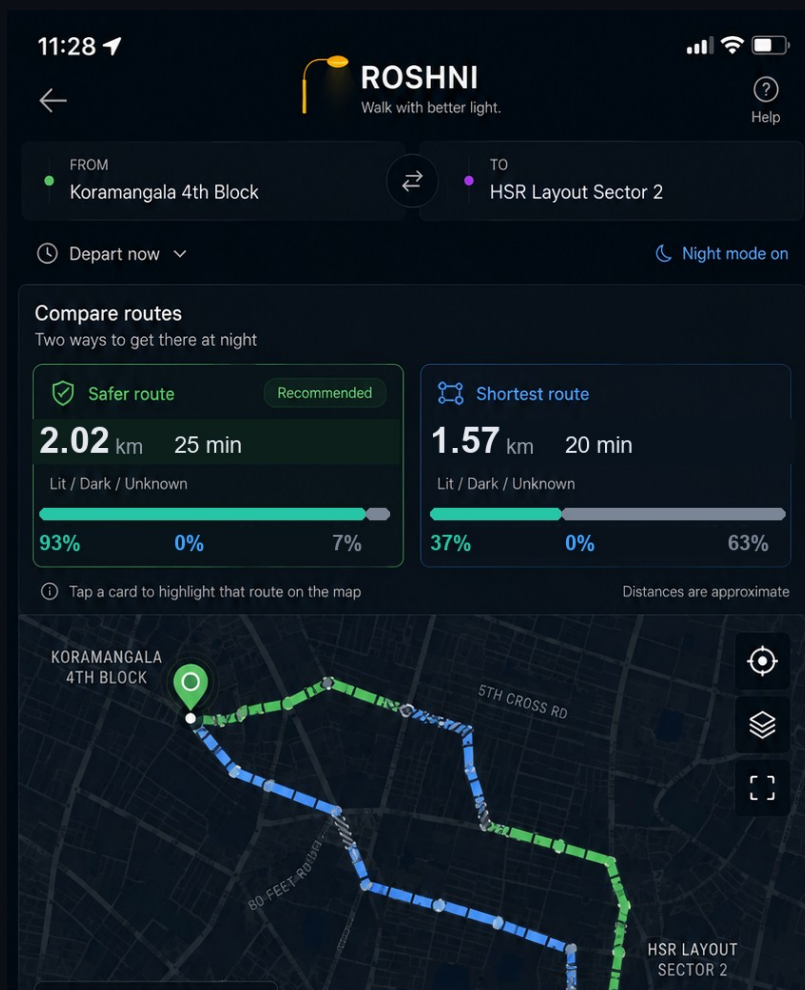
The municipal queue



The ratio, live — the budget slider recomputes in the browser. **Municipal priorities** — the municipality's to set, not ours. **Confidence is marked** — low-confidence rows are hatched, not hidden.

Prototype, running client-side on the fixture export. No network call on the demo path.

Both routes. Always.



Shortest: 1,575 m, and 63% of it crosses ground we are not confident about.

Safer: 2,021 m, 7% unknown — it routes around 18 low-confidence segments.

Six minutes to walk on ground we actually have data for.

The shortest route is always shown beside the safer one. We do not auto-select and we do not hide it — the choice is the walker's, and we say plainly which stretches we have no data for.

Routing cannot fix the worst streets

Bypass test on the 25 worst segments: delete the segment, look for another way between its endpoints.

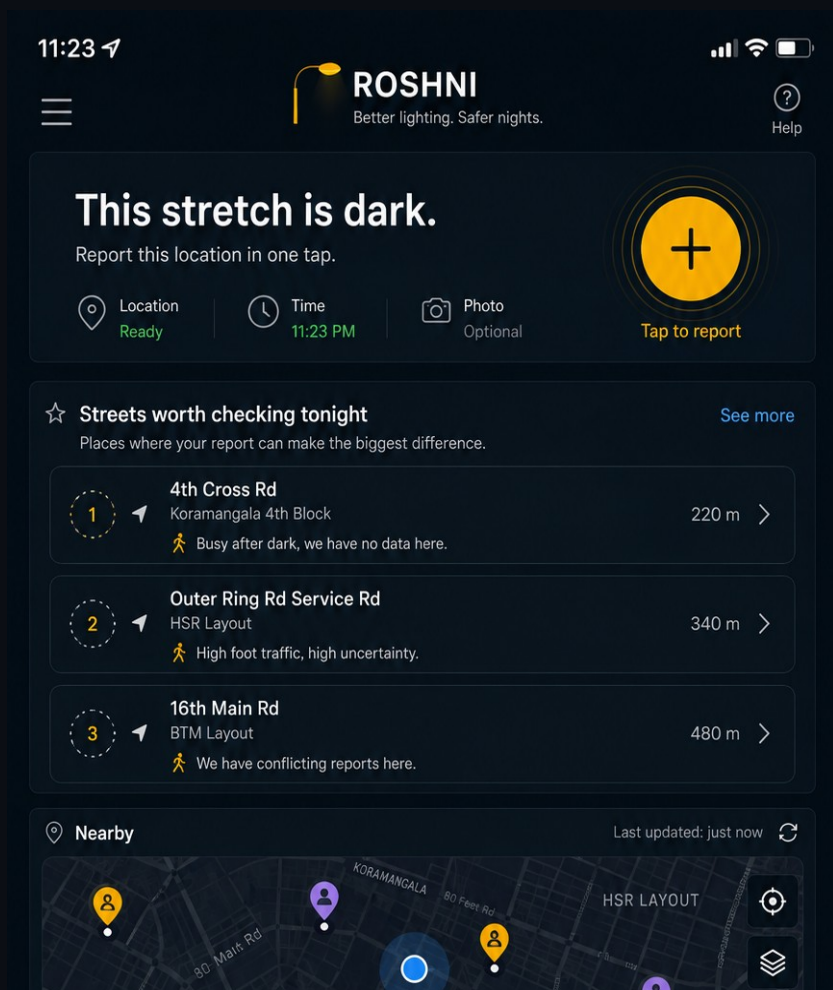
6 have no walkable alternative at all

19 have a bypass over 3× the segment's length

17.9× median bypass · worst 206.9×

On the streets that most need fixing there is nowhere else to walk. Routing cannot solve them. Only repair can.

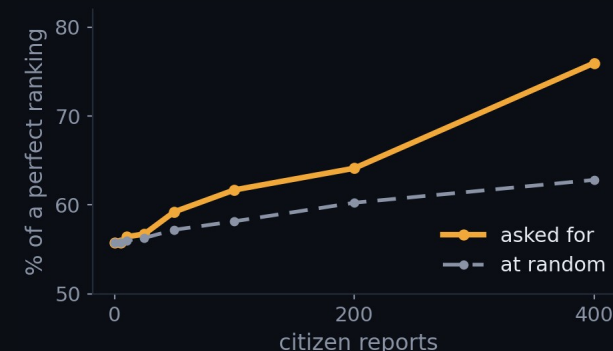
The empty state is the feature



The empty state is not “no reports yet.” It is **“streets worth checking tonight”** — so the app directs reporting instead of waiting for it.

93.7% of our 11,029 segments have no lamp evidence at all and run on the ward prior alone. That is not a flaw we are hiding — it is why every segment carries a variance, and it is what this screen is for.

A report is worth most exactly where we are least sure and most people walk. That is what this screen ranks by.

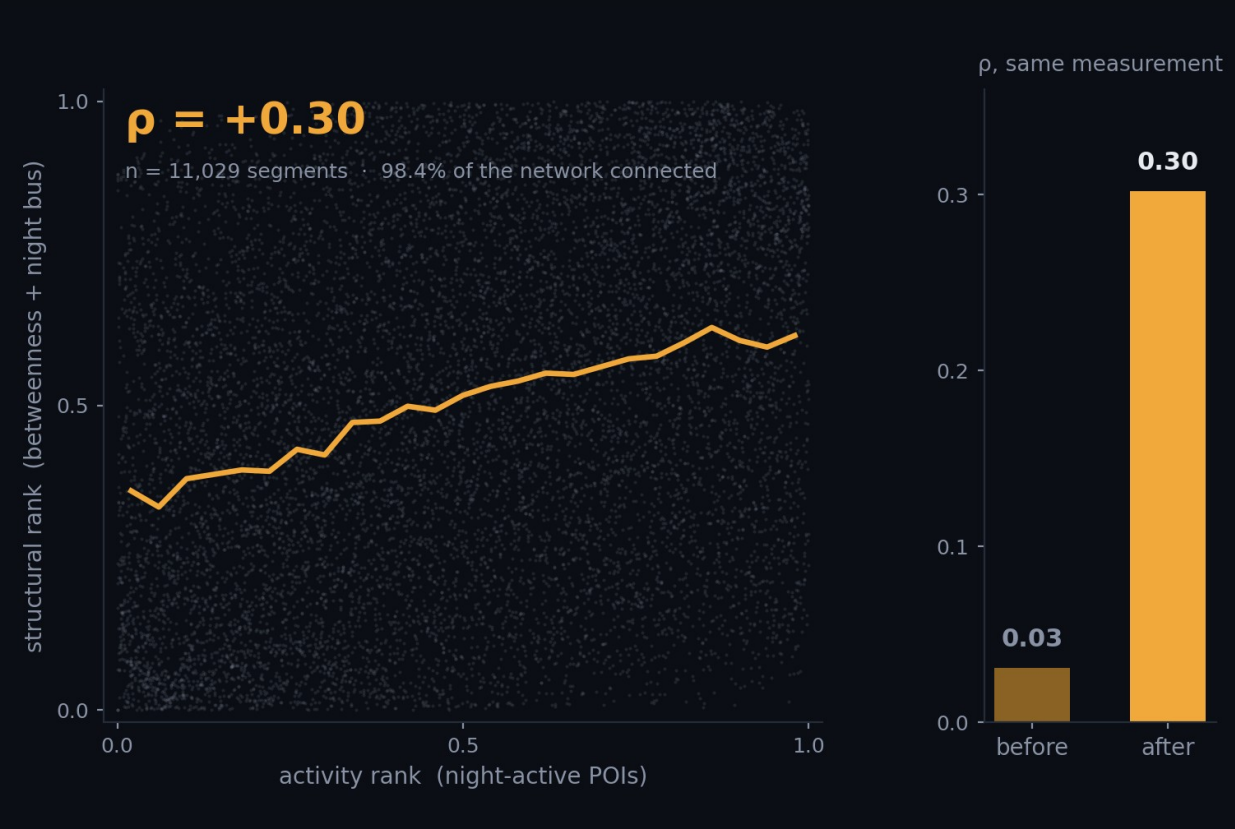


With zero reports the prior-only ranking already captures **56%** of what a perfectly-informed ranking would restore. 400 targeted reports takes it to **76%**; 400 random reports reach only 63%.

Simulation, not a field trial — labelled as one in out/curve.json.

How would you validate a footfall estimate with no official data?

Validation needs a signal you did not use to build the estimate. And you are validating an ordering, not a magnitude — “412 pedestrians an hour” is unfalsifiable; “these streets rank above those” is checkable.



We got a null first, and went looking for why

The first answer was $\rho = 0.03$. Two proxies that should have agreed didn't. So we went looking, and found we were keying the walk graph on **rounded coordinates** — junctions where two roads genuinely meet weren't merging. The network had shattered, and the betweenness we were feeding the model was computed over 55% of the city.

Keyed it on OSM node ids, re-ran over eight times the area. 98.4% now connects and $\rho = 0.30$ — and 0.44 against night bus departures.

Convergent	$\rho = +0.30$, and $+0.44$ vs night bus
Criterion — night pedestrian crashes	not yet measured
Ground truth — stratified field counts	not yet measured

The data exists. It's one approval away.

HAVE — OPEN, IN HAND

126,974 grievances for 2025 · six years available

1,508 mapped lamps · 1,128 labelled working / not working

Per-lamp KML for the ORR corridor and Bellandur/HSR

Lamp counts for all 198 wards — 421,113 lamps

BMTC GTFS with per-stop departure times

A two-date citizen lamp audit, 469 lamps

Night pedestrian crash reports

GATED — CATALOGUED, ONE APPROVAL AWAY

IUDX · Electronics City Industrial Township Authority

“Streetlights Location Info in Electronic City, Bengaluru” — with an hourly current-and-voltage companion. Marked Private. The sample record is public:

```
{"deviceId": "L0302",  
  "location": {"type": "Point",  
               "coordinates": [77.666046, 12.841758]}}
```

Roshni's lamp record is that record, copied verbatim. **“We plug into the municipal feed the day access is granted” is a schema fact, not a promise.** When it opens, the prior is replaced by measurement and the bands collapse. The queue, the exposure model, the knapsack and all three screens are unchanged.

Four things we get wrong, on purpose, out loud

1 93.7% of our segments run on a ward prior alone

690 of 11,029 carry any lamp evidence. Every other one carries a confidence band that says so — the router avoids the least-certain fifth of the network and the report screen targets it. But most of the map is an estimate, and we would rather say that than have you find it.

2 Our labelled lamps are geographically concentrated

OSM's working tags come from citizen surveys, and those surveys walked some wards and not others. We tested whether that flatters the result: restricted to surveyed wards it is 6.4×, with every label deleted 5.4×. It survives — but the concentration is real.

3 We cannot filter night activity the way we designed

OSM records opening_hours for 6.7% of POIs in our corridor — 78 of 1,157 — so we weight by POI class instead. And Google's review counts, which we do not have, undercount thelas and tea stalls: exactly the population most exposed to dark streets.

4 The working tags are survey-biased, and canopy is missing entirely

657 broken to 471 working is a property of who went out mapping, not Bengaluru's outage rate — we use them as labels, never as a base rate. Tree canopy is a large effect here and we have no data for it at all.

And one we are asked about: VIIRS nighttime lights are 500 m per pixel — five to ten city blocks — so they cannot tell you a street is dark. We do not use them that way.

What another week buys

Answered with what we would try to learn, not with features.

Close the criterion-validation leg

Mine the Bengaluru Traffic Police crash reports for the night pedestrian split. It is the only non-circular validator we have left.

Find out whether the ordering is right

Stratified field pedestrian counts at 21:30, and a rank correlation against what the model predicted.

Find out where the model stops transferring

Fit across wards of genuinely different urban form and report the degradation. Where it does not transfer, the system should refuse to guess and ask for a report — that behaviour is already built.

Ask “closed, or fixed?”

BBMP closes 96.4% of electrical complaints. Roshni is the first thing that can verify a closure with the person who reported it — a metric a commissioner does not have today.

First number they hear and the last.

THE ASK

The data exists, for our city, one approval away — and we already speak its schema.

Approve the IUDX request.

7.7×

Same crew. Same budget. Different order.

Appendix — every number, and where it came from

126,974	civic complaints, 1 Jan – 19 Jun 2025	1,508 / 1,128	lamps in snapshot / labelled working=yes no
39,847 · 31.4%	of them “Street Light Not Working”	690 · 6.3%	segments with any lamp evidence
96.4%	electrical closure rate (40,632 / 42,138)	27.6%	mean P(dark) across all segments
58.0%	road-maintenance closure rate	ρ +0.30 / +0.44	activity vs structural / vs night bus
421,113	streetlights across 198 BBMP wards	405 · 6,187	BMTC stops / departures after 21:00
11,029	segments scored · 1,180 km · 33 wards	3,417 / 9,199	night-active POIs of all POIs
95,557	walk-graph nodes · 98.4% one component	17.9×	median bypass, 25 worst segments

Counterfactual at a 40-repair budget: 4,544 vs 588 exposed pedestrian-km = 7.7×. Luckiest of 300 draws 4.9×. Steelman 1.9×. Surveyed wards only 6.4×. All lamp labels deleted 5.4×.

All figures out/stats.json, out/curve.json, out/routes.json — pipeline run 22 Aug 2026.